Zeshui Song, 2/14/26, HW3

1.9 part 2. (see HW 1. This problem will be graded) Using stress concentration factors, find the local maximum stresses at the pin joints in the tensile members. You will need to find bearing stress concentration factors for tight-fitting bearings (e.g online on amesweb) to solve this problem. Which stress (bearing stress or tensile stress) at which location(s) is most likely to be critical?

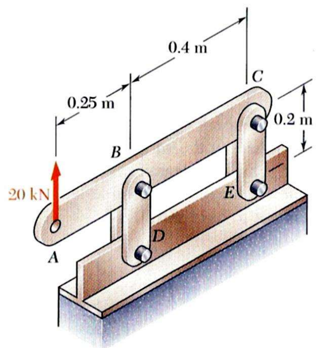

Fig. P1.9

Given (from HW 1 calculations):

- The 4 vertical links have a 8 x 36 mm rectangular cross section
- Each of the 4 pins have a 16 mm diameter
- Total force at BD (half of this acts on each link) $F_{BD} = 32.5\ kN$
- Calculated tensile stress in each link: $\sigma_{BD} = 101.56\ MPa$
- Calculated bearing stress in each link: $\sigma_b = 126.95\ MPa$

Find:

- Which stress and location are critical?

Approach:

- Using the stress concentration factor

$$K = \frac{\sigma_{max}}{\sigma_{ave}}$$

Using AMESWeb calculator, I can solve for the average normal tension and bearing stress concentration factors and apply it to find the critical location.

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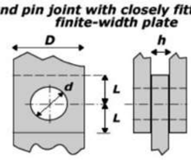
Inputs for calculation:

Round pin joint with closely fitting pin in finite-width plate



INPUT PARAMETERS		
Parameter	**Value**	
Plate width [D]	36	mm
Hole diameter [d]	16	mm
Hole center to plate edge distance [L]	36	mm
Plate thickness [h]	8	mm
Tension force [P]	16.25	kN

Calculate

Note: The calculator requires that $\frac{L}{D} \geq 1$**.** L, the distance from the hole center to the closest end, is not given nor is used in the stress concentration factor formula, so I assumed $L = 36\ mm$ to satisfy the condition. I think the reason for this condition is so that the link fails in a more predicable way (either from bearing or max normal tension), and not a combination.

Nominal stress based on net section:

$$\sigma_{na} = P/(D-d)h$$

Nominal stress based on bearing area:

$$\sigma_{nb} = P/dh$$

$$\sigma_{max\_a} = K_t \sigma_{na}$$

$$\sigma_{max\_b} = K_t \sigma_{nb}$$

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Results:

Parameter	Value	
TENSION STRESS		
Stress concentration factor [K_{ta}] *	2.65	---
Nominal tension stress based on net section [σ_{na}] °	101.56	MPa
Maximum tension stress [σ_{max}]	268.98	
BEARING STRESS		
Stress concentration factor [K_{tb}] **	2.19	---
Nominal bearing stress based on bearing area [σ_{nb}] x	126.95	MPa
Maximum bearing stress [σ_{max}]	277.61	

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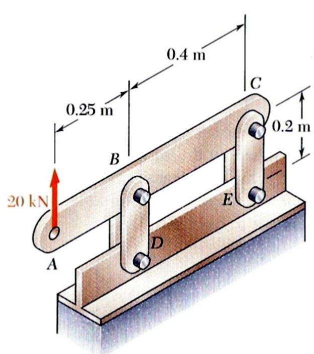

Fig. P1.9

The critical location is then at the inside of the holes in links BD, where the pin contacts the hole. This is because the maximum bearing stress is higher than the maximum normal tensile stress. If the link in tension were to fail, it would split the hole open along the direction of tension. [Time spent: 40 mins]

Note: The discussion could use some work. The two max stresses at B are quite similar. Is this a good design?
And note that link BD should extend at least one width beyond the bearing center at each end for this to be a valid analysis.

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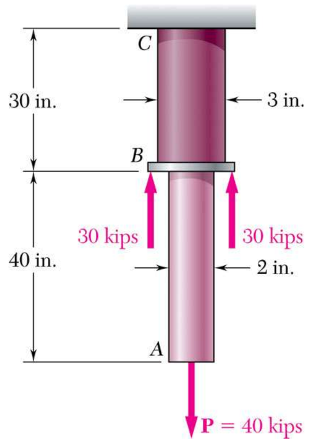
2.19 Two solid cylindrical rods are joined at B and loaded as shown. Rod AB is made of steel ($E = 29 \times 10^6$ psi), and rod BC of brass ($E = 15 \times 10^6$ psi). Determine (*a*) the total deformation of the composite rod ABC, (*b*) the deflection of point B.



Given:
- $P = 40\ kips, F = 30\ kips$
- Rod AB:
  - $d_{AB} = 2\ in$
  - $E_{steel} = 29 \times 10^6\ psi$
  - $L_{AB} = 40\ in$
- Rod BC:
  - $d_{BC} = 3\ in$
  - $E_{brass} = 15 \times 10^6\ psi$
  - $L_{BC} = 30\ in$

Find:
- Total deformation of the rod ABC
- Deflection of point B

Approach:
- Applying statics to find the internal reaction forces in rods AB and BC
- Find deflection using

$$\delta = \frac{PL}{EA}$$

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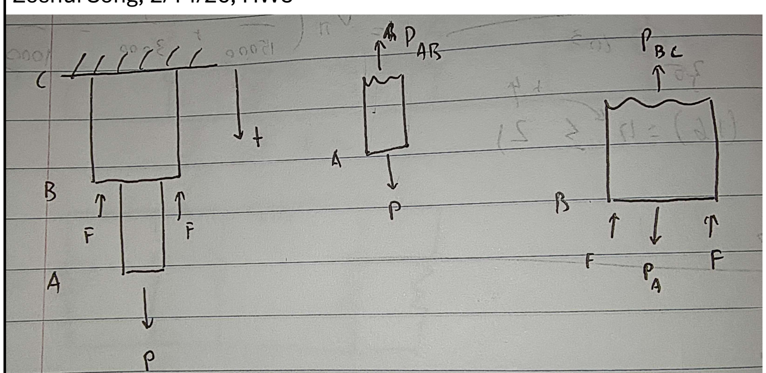

$$\sum F_y = 0 = P - P_{AB} \rightarrow P_{AB} = P$$

$$\sum F_y = 0 = P_A - 2F - P_{BC} \rightarrow P_{BC} = P_A - 2F$$

$$P_{BC} = P - 2F$$

$$\delta_{AB} = \frac{P_{AB}L_{AB}}{E_{steel}A_{AB}} = \frac{[40\ kips]\left[\frac{10^3\ lbf}{1\ kip}\right] \cdot [40\ in]}{[29 \times 10^6\ psi\,] \cdot \frac{1}{4}\pi[2\ in]^2} = 0.01756\ in$$

$$\delta_{BC} = \frac{P_{BC}L_{BC}}{E_{brass}A_{BC}} = \frac{[40 - 2 \cdot 30\ kips]\left[\frac{10^3\ lbf}{1\ kip}\right] \cdot [30\ in]}{[15 \times 10^6\ psi] \cdot \frac{1}{4}\pi[3\ in]^2} = -0.005658\ in$$

$$\boxed{\delta_{Tot} = \delta_{AB} + \delta_{BC} = 0.0119\ in}$$

$$\boxed{\delta_B = \delta_{BC} = -0.005658\ in}$$

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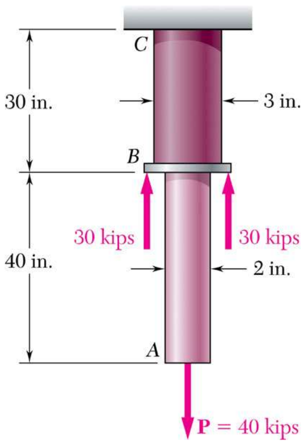

$$\boxed{\delta_{Tot} = \delta_{AB} + \delta_{BC} = 0.0119\ in}$$

$$\boxed{\delta_B = \delta_{BC} = -0.005658\ in}$$

Rod BC gets compressed since because the net applied load is upwards, while rod AB gets stretched. [Time spent: 20 mins]

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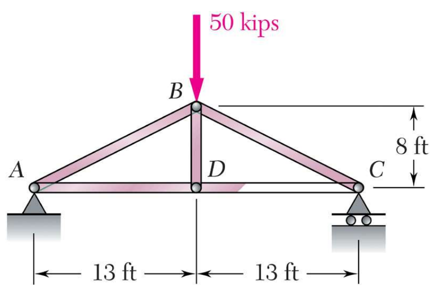
Problem 2.21 will be graded.

2.21 For the structural steel (ASTM-A36) truss and loading shown, determine the deformations of members *AB* and *AD*, which have cross-sectional areas of 4.0 in^2 and 2.8 in^2, respectively. You may assume that all members are stable (ie no buckling occurs), and this assumption should be stated briefly in your discussion as a limitation in your analysis (along with other limitations).



Given:
- ASTM-A36: $E = 29 \times 10^6\ psi$
- $A_{AB} = 4\ in^2$
- $A_{AD} = 2.8\ in^2$
- All other members (BD, BC, DC) has no deformation

Find:
- Deformations of members AB and AD

Approach:
- Since these are all 2 force members, I can solve for the internal forces using statics
- Find deflection using

$$\delta = \frac{PL}{EA}$$

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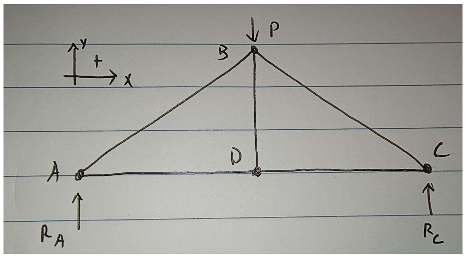
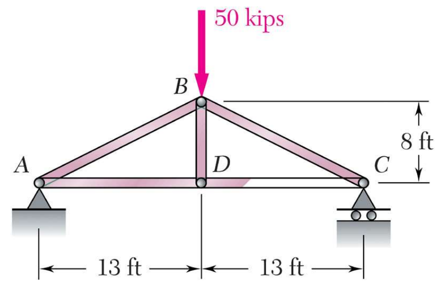
Full structure FBD:

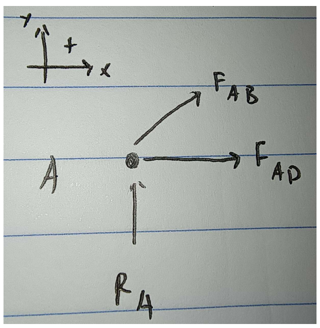


$$\sum M_A = 0 = R_C[36\,ft] - P[13\,ft] \rightarrow R_C = \frac{P}{2}$$

$$\sum F_y = 0 = R_A + R_C - P \rightarrow R_A = R_C = \frac{P}{2}$$



FBD at pin A



$$\sum F_x = 0 = F_{AB}\frac{13}{\sqrt{13^2+8^2}} + F_{AD}$$

$$\sum F_y = 0 = R_A + F_{AB}\frac{8}{\sqrt{13^2+8^2}}$$

$$\downarrow$$

$$F_{AB} = \frac{\sqrt{13^2+8^2}}{8}R_A = \frac{\sqrt{13^2+8^2}}{16}P \qquad F_{AD} = -\frac{13}{16}P$$

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$$\boxed{F_{AB} = \frac{\sqrt{13^2+8^2}}{16}P = 47.701\,kips} \qquad \boxed{F_{AD} = -\frac{13}{16}P = -40.625\,kips}$$

Note: The geometry is a bit inaccurate (AB is not 16' long) otherwise good writeup. The discussion could use some work again, especially a warning that buckling should be considered in compression members, since it could significantly increase deflection. It is interesting to note that the downward deflections of B and D are quite a bit larger than the axial deflections of the members, and it might be interesting to estimate this ...

$$\delta_{AB} = \frac{F_{AB}L_{AB}}{EA_{AB}} = \frac{[47.701\,kips]\left[\frac{10^3\,lbf}{1\,kip}\right]\cdot\left[\sqrt{13^2+8^2}\,ft\right]}{[29\times10^6\,psi]\cdot[4\,in^2]} = 0.006276\,ft$$

$$\delta_{AD} = \frac{F_{AD}L_{AD}}{EA_{AD}} = \frac{[-40.625\,kips]\left[\frac{10^3\,lbf}{1\,kip}\right]\cdot[13\,ft]}{[29\times10^6\,psi]\cdot[2.8\,in^2]} = -0.0062504\,ft$$

$$\boxed{\delta_{AB} = 0.006276\,ft}$$

$$\boxed{\delta_{AD} = -0.0062504\,ft}$$

Since we assumed only AB and AD can be deformed, the whole truss structure will twist in the clockwise direction as AB stretches and AD compresses. If we allowed the other members to also deform, then the total deformation will be much higher. [Time spent: 20 mins]

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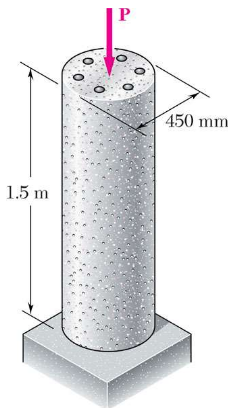
2.37 The 1.5-m concrete post is reinforced with six steel bars, each with a 28-mm diameter. Knowing that $E_s = 200$ GPa and $E_c = 25$ GPa, determine the normal stresses in the steel and in the concrete when a 1550 kN axial centric force **P** is applied to the post.



Given:
- $P = 1550\ kN$
- $L = 1.5\ m$
- $E_s = 200\ GPa$
- $E_c = 25\ GPa$
- $d_s = 28\ mm$
- There are SIX steel bars
- $d_c = 450\ mm$

Find:
- Normal stress in steel and concrete

Approach:
- Since the steel and concrete are bonded, they must have the same deflection under load P

$$\delta = \frac{PL}{EA}$$

- Using the deflection constraint and statics, I can then find the loads on the concrete and steel separately and then solve for stress

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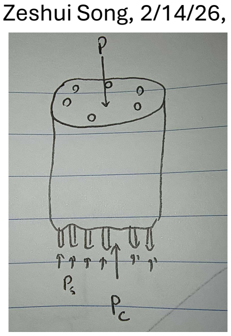

$$\sum F = 0 = P_c + P_s - P \rightarrow P = P_c + P_s$$

$$\delta = \frac{P_c L}{E_c A_c}$$

$$\delta = \frac{P_s L}{E_s A_s}$$

$$\frac{P_c L}{E_c A_c} = \frac{P_s L}{E_s A_s} \rightarrow P_c = P_s \frac{E_c A_c}{E_s A_s}$$

$$A_c = \frac{1}{4}\pi d_c^2 - \frac{6}{4}\pi d_s^2 = 155348.61\ mm^2$$

$$A_s = \frac{6}{4}\pi d_s^2 = 3694.512\ mm^2$$

$$P_c = 5.25605\ P_s$$

Plug into force balance:

$$P = P_c + P_s$$

$$P = 5.25605\ P_s + P_s \rightarrow P_s = 0.1598P \qquad \boxed{P_s = 247.759\ kN} \qquad \boxed{P_c = 1302.238\ kN}$$

Stress:

$$\boxed{\sigma_s = \frac{P_s}{A_s} = \frac{[247.759\ kN]\left[\frac{10^3\ N}{kN}\right]}{[3694.512\ mm^2]\left[\frac{(10^{-3}\ m)^2}{mm^2}\right]} = 67.06 \times 10^6\ Pa}$$

$$\boxed{\sigma_c = \frac{P_c}{A_c} = \frac{[1302.238 kN]\left[\frac{10^3\ N}{kN}\right]}{[155348.61\ mm^2]\left[\frac{(10^{-3}\ m)^2}{mm^2}\right]} = 8.382 \times 10^6\ Pa}$$

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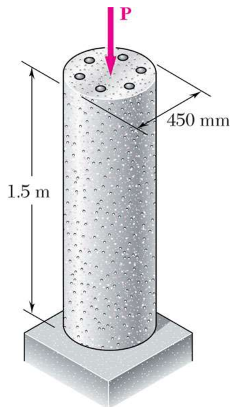

$\boxed{\sigma_s = 67.06 \times 10^6\ Pa}$

$\boxed{\sigma_c = 8.382 \times 10^6\ Pa}$

It turns out the concrete takes about 5 times more load than the steel rod, but the steel is under more stress due to its smaller surface area. [Time spent: 30 mins]

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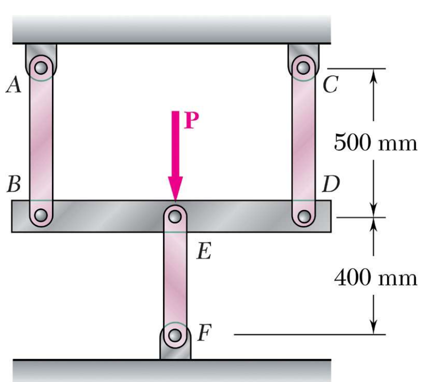
2.41 Three steel rods (E = 200 GPa) support a 36-kN load **P**. Each of the rods AB and CD has a 200-mm^2 cross-sectional area and rod EF has a 625-mm^2 cross-sectional area. Determine (a) the change in length of rod EF, (b) the stress in each rod.



Given:
- $P = 36\ kN$
- $E = 200\ GPa$
- $A = A_{AB} = A_{CD} = 200\ mm^2$
- $A_{EF} = 625\ mm^2$

Find:
- Deflection of EF
- Stress in each rod

Approach:
- Since links AB and CD have equal lengths, areas, material, and deflection, the loads they see must also be equal

$$\delta = \frac{PL}{EA}$$

- I will simplify links AB and CD into 1 link with the same net load and deflection P_T, δ_T (and the sum of the cross-sectional areas)
- The total height of this set up is fixed so the total deflection is zero

$$\delta_T + \delta_{EF} = 0$$

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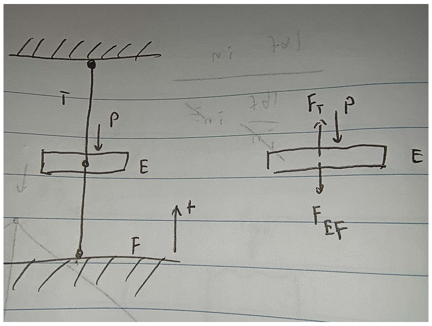

$$\sum F = 0 = F_T - F_{EF} - P \rightarrow P = F_T - F_{EF}$$

$$\delta_T = \frac{F_T L_T}{E(2A)}$$

$$\delta_{EF} = \frac{F_{EF} L_{EF}}{E A_{EF}}$$

$$\delta_T = -\delta_{EF}$$

$$\frac{F_T L_T}{E(2A)} = -\frac{F_{EF} L_{EF}}{E A_{EF}}$$

$$F_T = -F_{EF} \frac{L_{EF}}{A_{EF}} \frac{2A}{L_T} = -0.512 F_{EF}$$

Plug into force balance:

$P = -0.512 F_{EF} - F_{EF} \rightarrow F_{EF} = -0.6613P = -23.809\ kN$

$F_T = -0.512 F_{EF} = 12.19\ kN$

Thus, the loads on each member is

$F_{AB} = F_{CD} = \frac{F_T}{2} = 6.095\ kN \qquad F_{EF} = -23.809kN$

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Solve for stress and deflection:

$F_{AB} = F_{CD} = \frac{F_T}{2} = 6.095\ kN \qquad F_{EF} = -23.809kN$

$$\delta_{EF} = \frac{F_{EF} L_{EF}}{E A_{EF}} = \frac{[-23.809kN]\left[\frac{10^3 N}{kN}\right] \cdot [400\ mm]\left[\frac{10^{-3} m}{mm}\right]}{[200\ GPa]\left[\frac{10^9\ Pa}{GPa}\right] \cdot [625\ mm^2]\left[\frac{(10^{-3} m)^2}{mm^2}\right]} = -7.618 \times 10^{-5}\ m$$

$$\sigma_{AB} = \sigma_{CD} = \frac{F_{AB}}{A} = \frac{[6.095kN]\left[\frac{10^3 N}{kN}\right]}{[200\ mm^2]\left[\frac{(10^{-3} m)^2}{mm^2}\right]} = 3.047 \times 10^7\ Pa$$

$$\sigma_{EF} = \frac{F_{EF}}{A_{EF}} = \frac{[-23.809kN]\left[\frac{10^3 N}{kN}\right]}{[625mm^2]\left[\frac{(10^{-3} m)^2}{mm^2}\right]} = -3.809 \times 10^7\ Pa$$

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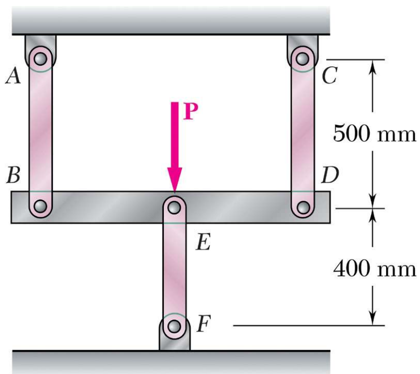

$\boxed{\delta_{EF} = -7.618 \times 10^{-5}\ m}$

$\boxed{\sigma_{AB} = \sigma_{CD} = 3.047 \times 10^{7}\ Pa}$ $\boxed{\sigma_{EF} = -3.809 \times 10^{7}\ Pa}$

This makes sense since for the total height to be constant, one side of E has to be compressed while the other is stretched. The stresses on the members are roughly equal in terms of magnitude while EF is in compression and AB & CD are in tension. [Time spent: 40 mins]

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Zeshui Song, 2/14/26, HW3

2.51 A steel railroad track ($E_s = 29 \times 10^6$ psi, $\alpha_s = 6.5 \times 10^{-6}$/°F) was laid out at a temperature of 30°F. Determine the normal stress in the rails when the temperature reaches 125°F, assuming that the rails (*a*) are welded to form a continuous track, (*b*) are 39 ft long with $\frac{1}{4}$-in. gaps between them.

Given:
- $E_s = 29 \times 10^6\ psi$
- $\alpha_s = 6.5 \times 10^{-6}/°F$
- $T_i = 30°F$

Find:
- Normal stress in the rail when $T_f = 125°F$
  - When the rails are continuous
  - When the rails are 39 ft long with 0.25 in gaps

Approach:
- For (a) we can used the constrained thermal stress equation
$$\sigma_T = \alpha \Delta T E$$
- For (b), there will be zero stress until the rails expand past 0.25 in. The amount of deflection beyond 0.25 in can be converted to stress by rearranging the deflection formula
$$\delta = \frac{PL}{EA} \rightarrow \sigma = \frac{P}{A} = \frac{\delta E}{L}$$

(a) $\sigma_T = \alpha \Delta T E = [6.5 \times 10^{-6}][125 - 30°F][29 \times 10^6\ psi]$

$\boxed{\sigma_T = 17907.5\ psi\ (compressive)}$

(b) $\delta_T = \alpha \Delta T L = [6.5 \times 10^{-6}][125 - 30°F][39\ ft]\left[\frac{12\ in}{ft}\right] = 0.28899\ in$

$\delta = [0.28899\ in] - [0.25\ in] = 0.03899\ in$

$$\sigma = \frac{[0.03899\ in] \cdot [29 \times 10^6\ psi]}{[39\ ft]\left[\frac{12\ in}{ft}\right]} = 2416.047\ psi$$

$\boxed{\sigma = 2416.047\ psi\ (compressive)}$

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This result makes sense since the fully constrained rail in (a) has larger normal stress while the one with gaps in (b) has much lower internal stress. [Time spent: 20 mins]

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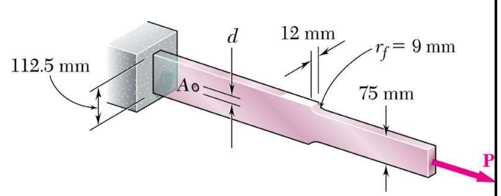
2.99 A hole is to be drilled in the plate at A. The diameters of the bits available to drill the hole range from 12 to 24 mm in 3-mm increments. (*a*) Determine the diameter d of the largest bit that can be used if the allowable load at the hole is to exceed that at the fillets. (*b*) If the allowable stress in the plate is 145 MPa, what is the corresponding allowable load **P?** (*c*) For your design for d, estimate the one-time maximum load that the plate could withstand at failure. Use the bilinear stress-strain model (ie assume that the stress during plastic deformation is equal to the yield stress). The plate is constructed from ASTM-A992 steel.

Approach:
- (a) Start by finding the max load (using stress concentration factor K_{fillet}) at the fillet cross-section given allowed stress. Then solve for max loads across varying diameters.
- (b) With the diameter d determined, solve for the maximum allowable load. Because d was chosen to be stronger than the fillet, this maximum load is equal to the maximum load at the fillet found in part (a).
- (c) Instead of using the allowed stress, we will use the yield stress of the steel to find the max load at the fillet. (since this is during plastic deformation, the stress concentration factor will not apply)

Given:
- $d = 12, 15, 18, \ldots mm\ to\ 24\ mm$
- $E = 200\ GPa$
- $w = 112.5\ mm$
- $t = 12\ mm$
- Allowed stress: 145 Mpa
- $\sigma_{yield} = 345\ MPa$

Find:
- (a) The largest diameter d so that the max load at hole is larger than at fillet (fillet is weaker)
- (b) What is the allowable load for this bar?
- (c) For d, estimate one-time max load at failure.



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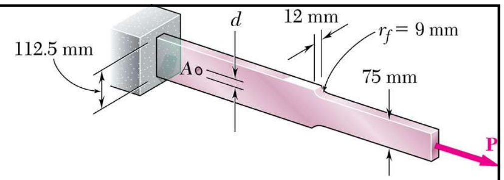
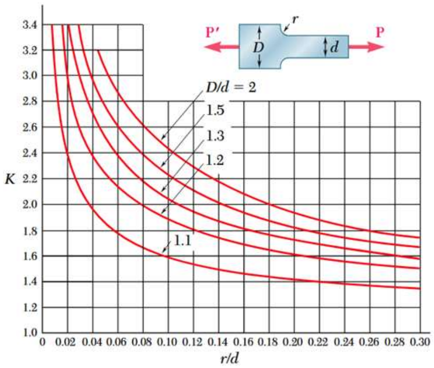


(a) $\sigma_{max,fillet} = K_{fillet}\sigma_{fillet} = K_{fillet}\dfrac{P_{all}}{A_{fillet}}$



$$\frac{r}{d} = \frac{[9\ mm]}{[75\ mm]} = 0.12$$

$$\frac{D}{d} = \frac{[112.5\ mm]}{[75\ mm]} = 1.5$$

$\rightarrow$ $K_{fillet} = 2.1$

The max stress at fillet will be equal to allowable stress

$$P_{all} = \frac{\sigma_{all}A_{fillet}}{K_{fillet}} = \frac{[145 \times 10^6 Pa]\cdot[75\cdot 12\ mm^2]\left[\frac{(10^{-3}m)^2}{mm^2}\right]}{2.1} = 62142.85\ N$$

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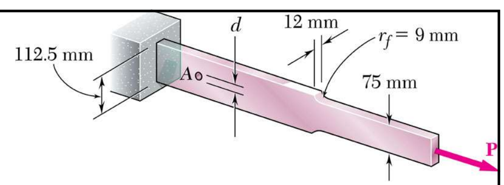
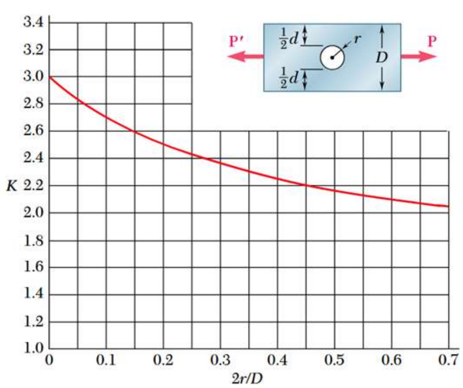


(a) $\sigma_{max,hole} = K_{hole}\sigma_{hole} = K_{hole}\dfrac{P_{all}}{A_{hole}}$

The max stress at hole will also be equal to allowable stress

$$P_{all} = \frac{\sigma_{all}A_{hole}}{K_{hole}} = \frac{\sigma_{all}\cdot(w-d)t}{K_{hole}}$$



d=2r	D	d/D	K_hole
12	112.5	0.10666667	2.7
15	112.5	0.13333333	2.64
18	112.5	0.16	2.59
21	112.5	0.18666667	2.53
24	112.5	0.21333333	2.48

Checked with AmesWeb Calculator

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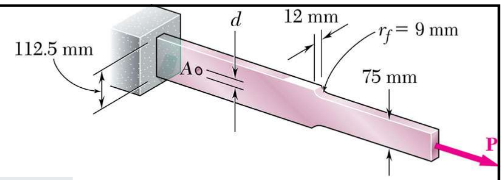

(a) Solving for the max allowable load is then

$$P_{all} = \frac{\sigma_{all} A_{hole}}{K_{hole}} = \frac{\sigma_{all} \cdot (w-d)t}{K_{hole}}$$

d [mm]	sigma_all [Pa]	(w-d)t [m^2]	K_hole	Allowed load [N]
12	145000000	0.001206	2.7	64766.66667
15	145000000	0.00117	2.64	64261.36364
18	145000000	0.001134	2.59	63486.48649
21	145000000	0.001098	2.53	62928.85375
24	145000000	0.001062	2.48	62092.74194

The max diameter is then 21 mm

(b) $\boxed{62142.85\ N \text{ (from part a)}}$

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(c)

$$\sigma_{yield} = \frac{P_{max}}{A_{min}}$$

The min area is again at the fillet area, since the width of (w-d) is still larger

$$\boxed{P_{max} = [345 \times 10^6 Pa] \cdot [75 \cdot 12\ mm^2] \left[\frac{(10^{-3} m)^2}{mm^2}\right] = 310500\ N}$$

This makes sense that the max loading at failure will be larger than the allowable loads. Throughout the problem, we have seen that the failure point is at the fillet section since the we designed the hole to be stronger. [Time spent: 50 mins]

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